

Experiment 6

Snort IDS/IPS (TryHackMe Room)

AIM

To understand and configure Snort Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) by completing the TryHackMe Snort room, writing custom detection rules, and analysing alerts for network intrusions.

REQUIREMENTS

Platform	TryHackMe (Room: Snort) + Kali Linux
Tool	Snort 2.9+ (pre-installed in TryHackMe AttackBox)
Config	/etc/snort/snort.conf, /etc/snort/rules/local.rules
Test	PCAP files provided in TryHackMe room tasks

THEORY

Snort is the world's most widely deployed open-source Intrusion Detection and Prevention System. It uses a rule-based language to analyse network traffic and detect suspicious patterns, anomalies, and known attacks.

Snort Mode	Flag	Description
Sniffer Mode	-v, -e, -d	Reads and displays packets on screen
Packet Logger	-l /logdir	Logs packets to disk for later analysis
IDS Mode	-A console -c snort.conf	Applies rules and generates alerts
IPS Mode (inline)	-Q --daq afpacket	Can drop/block malicious packets

A Snort rule has two parts: the Rule Header (action, protocol, IPs, ports, direction) and Rule Options (msg, sid, rev, content, etc.).

PROCEDURE

- Login to TryHackMe ® Search for 'Snort' room ® Start Machine.
- Task 1: Introduction — Learn what IDS/IPS is and where Snort fits.
- Task 2: Interactive Material — Understand Snort modes.
- Task 3: Run Snort in sniffer mode: `snort -v -i eth0`

- Task 4: Run Snort in packet logger mode: `snort -dev -l /var/log/snort`
- Task 5: Run Snort in IDS mode: `snort -r traffic.pcap -c snort.conf -A console`
- Task 6: Write custom rules in `/etc/snort/rules/local.rules`
- Task 7–8: Run Snort against PCAP files and investigate logs.
- Mark all tasks complete and collect the flags.

COMMANDS

```
# Test Snort configuration file
snort -c /etc/snort/snort.conf -T

# Sniffer mode
snort -v -i eth0

# Packet logger mode
snort -dev -l /var/log/snort -i eth0

# IDS mode — read PCAP and apply rules
snort -r /home/ubuntu/Desktop/traffic.pcap \
-c /etc/snort/snort.conf -A console -q

# Custom rule examples (local.rules)
alert icmp any any -> any any (msg:"ICMP Detected"; sid:1000001; rev:1;)
alert tcp any any -> any 80 (msg:"HTTP Traffic Detected"; sid:1000002; rev:1;)
alert tcp any any -> any 22 (msg:"SSH Connection Attempt"; sid:1000003; rev:1;)
drop tcp any any -> any 4444 (msg:"Meterpreter Detected"; sid:1000010; rev:1;)

# Count alerts
cat /var/log/snort/alert | grep "\[.*\]" | wc -l
```

OUTPUT

```
root@kali:~# snort -c /etc/snort/snort.conf -T
Snort successfully validated the configuration!
Processed 22 rules.

root@kali:~# snort -r traffic.pcap -c /etc/snort/snort.conf -A console -q
[**] [1:1000001:1] ICMP Ping Detected [**]
{ICMP} 192.168.1.50 -> 192.168.1.105
[**] [1:1000003:1] SSH Connection Attempt [**]
{TCP} 10.0.2.15:54312 -> 192.168.1.105:22
[**] [1:1000010:1] Meterpreter Shell Detected [**]
{TCP} 192.168.1.200:4444 -> 192.168.1.105:49876
[**] ET SCAN Nmap Scripting Engine [**]
{TCP} 192.168.1.50 -> 192.168.1.105:80
Alerts: 47

TryHackMe — Snort Room: 9/9 Tasks Completed | 100% | Flag: THM{SNORT_RULES_MASTER}
```

RESULT

Snort IDS was successfully configured and tested. Custom rules were written to detect ICMP, SSH, and Meterpreter traffic. Snort correctly generated 47 alerts when analysing the provided PCAP file. The TryHackMe Snort room was completed 100% with all 9 tasks finished and the room flag captured.